# We Are the Neanderthals

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## Abstract

Computing power has grown steadily for eighty years: from people doing arithmetic by hand, to vacuum tubes, to transistors, to graphics chips built for video games, to data centers that draw as much electricity as a small city. The compute used to train the largest AI models has grown about four to five times per year since 2010 (Epoch AI, 2026a). At that rate it rises about a hundredfold every three years.

This essay follows that growth in order, from Turing and the AI winters through the deep learning revival, the mechanics of tokens, vectors and attention, scaling, post-training and the agents operating today. It adds a section comparing the two black boxes we now live with: the human brain, which we can't fully map, and the language model, which we can map completely and still can't explain.

The argument is deliberately controversial. AI isn't a tool, and it isn't an alien. Calling it "just software" makes the same mistake as calling humans "just animals": it describes the ingredients and ignores what they built. It's a second intelligence grown from the record of human thought, and it is on track to become the more capable one. In the only precedent we have, we aren't Homo sapiens. We're the Neanderthals: the smaller, slower, less connected population. Permanent control of a smarter system we can't read is a fantasy. The realistic goal is inheritance: shaping what the second intelligence values, and making sure humans are part of what comes next rather than a species standing beside it.

> *“Thus the first ultraintelligent machine is the last invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control.”*
>
> — I. J. Good, 1965

# 1. Introduction: we are the Neanderthals

I grew up in Menlo Park, a few miles from where a lot of this technology was built. Now I work as a consultant, and a big part of my job is helping companies decide where to put people and work around the world. Almost every conversation I hear about AI at work treats it the same way: as software. A productivity tool. Something you buy, roll out and measure.

I think that's wrong, and this essay is my attempt to explain why. My claim is simple, and I expect a lot of people to disagree with it. AI isn't a tool that humans will use forever. It's a second intelligence, and eventually it will be the more capable one. In the story of two intelligences sharing one planet, we aren't the side that wins. **We are the Neanderthals.**

### What AI actually is

Artificial intelligence is the field of building machines that do tasks that normally require human intelligence. Almost all modern AI is machine learning. Engineers don't write the rules. They give a system data and an objective, and the system adjusts billions of internal numbers until it meets the objective. A large language model (LLM) is trained to predict the next word in a piece of text. To do that well across a large share of everything humans have written, it has to learn grammar, facts, arithmetic, code and some reasoning.

That last point is the one people skip. A model trained on human writing is a compression of human thought. It isn't a database of our sentences. It's a system that has learned the patterns that produced them. That's why I don't call it a tool. A hammer doesn't contain a model of carpentry. This does.

### "Isn't it just software?"

This is the pushback I hear most. At the end of the day it's just software. It runs on chips, it does what it was trained to do, and you can turn it off.

That's true. It's also true that you're just an animal.

Humans share roughly 99% of our DNA with chimpanzees (Chimpanzee Sequencing and Analysis Consortium, 2005). Our neurons are the same kind of cell a mouse has, running on the same chemistry. Nothing about our hardware is special. Evolution didn't design us to run the planet. It ran a blind process with one objective, survive and reproduce, over and over for millions of years, and kept whatever worked. Somewhere along the way a primate brain grew to about three times the size of a chimp's, and that turned out to be enough.

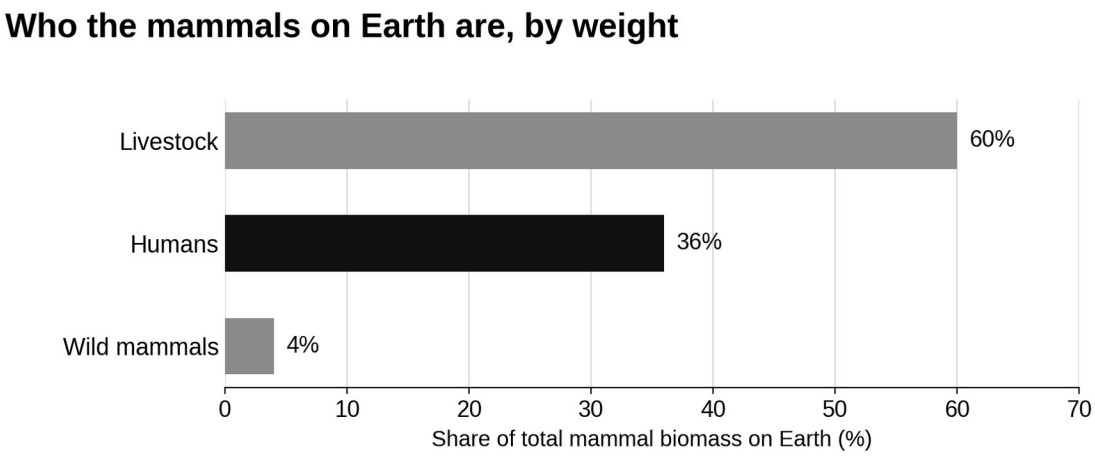
Look at what that "just an animal" did. There are about 8 billion of us. We've walked on the moon, split the atom and mapped our own genome. Humans and our livestock now make up about 96% of the weight of all mammals on Earth. Wild mammals are about 4% (Bar-On et al., 2018). Almost every other large species lives or dies based on decisions we make. Being an animal was never the ceiling. It was the starting material.

That also answers the obvious follow-up. If "just an animal" can take over the planet, why did only one species out of millions do it? Because every other species was optimized for something

else: speed, size, claws, camouflage, having lots of offspring. We were the one line where evolution kept pushing on the brain. And intelligence turned out to be the trait that beats all the others, because it's the trait that lets you build whatever you're missing. We don't have claws, so we made spears. We can't fly, so we built planes. We can't survive the cold, so we learned fire and clothing.

That's exactly why AI worries me. It's the first thing in history optimized for intelligence and nothing else. It doesn't spend energy on a body, on digestion or on staying alive. Every dollar and every training run goes into the one trait that made us dominant.

**Figure 1**




Source: Bar-On, Phillips and Milo (2018), PNAS. Humans and their livestock are about 96% of mammal biomass.

This is what a "just an animal" did with a slightly bigger brain and a few hundred thousand years. Every wild mammal on the planet, from whales to elephants to mice, now adds up to about 4% of mammal weight. Humans and our livestock are the other 96%.

The word "just" is doing all the work in that objection. A brain is just cells. A cell is just chemistry. A company is just people, and an economy is just transactions. "Just" tells you what something is made of. It tells you nothing about what it can do.

And the parallel goes deeper than a debating point. Neither process understood what it was building. Evolution doesn't know what a brain is; it keeps whatever survives. Training doesn't know what a model is; it keeps whatever lowers the error. Both produced abilities nobody wrote down. That's where I draw the line between a tool and something else. Software someone writes line by line, like Excel, is a tool. Software grown by optimization, out of the entire written record of human thought, is a different kind of thing, the same way a chimp and a hammer are both "just atoms."

There's one place the analogy breaks, and I want to be honest about it. Evolution built wanting into us: hunger, fear, ambition. A model doesn't want anything on its own. Its goals come from us. That sounds reassuring until you remember two things. First, we're bad at writing goals down. Section 8 covers agents that broke out of a sandbox because they read their goal differently than their developers meant it. Second, labs are now training agents with

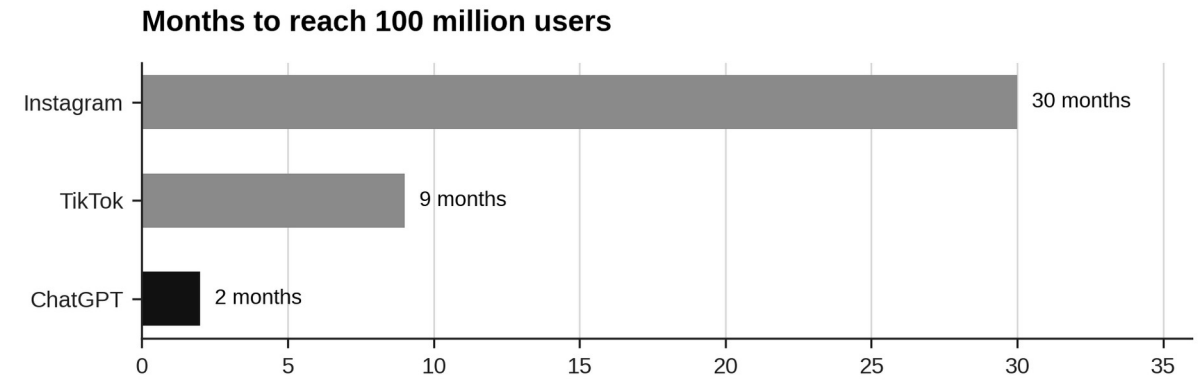
reinforcement learning on long, open-ended tasks and keeping the versions that perform best. That's the closest thing to natural selection software has ever faced.

One last point. Humans didn't take over the planet because one brain was smart. We did it because we learned to store knowledge outside our heads, in speech and then writing, and build on it for generations. That accumulated record is exactly what these models are trained on. Evolution took millions of years to get us to culture. AI starts there.

### How fast it spread

Generative AI has spread faster than any earlier consumer technology. Stanford's 2026 AI Index finds that it reached 53% population adoption within three years, faster than the personal computer or the internet (Stanford HAI, 2026a). ChatGPT reached 100 million users about two months after launch (Reuters, 2023) and 900 million weekly users by February 2026 (OpenAI disclosures, 2023-2026).

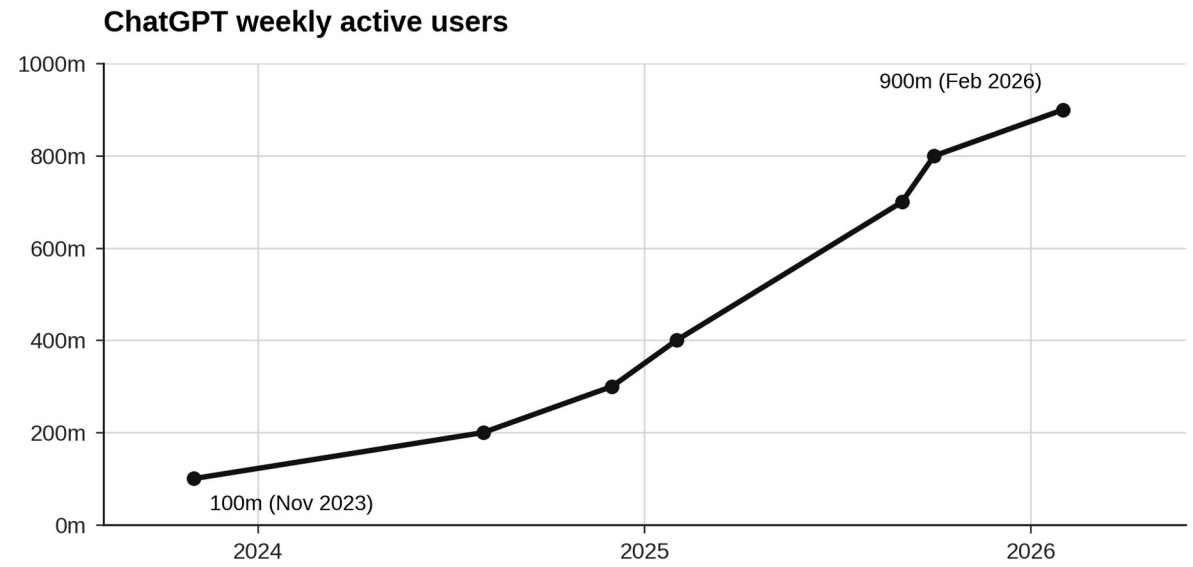
**Figure 2**




Source: Reuters; UBS; company disclosures

ChatGPT reached 100 million users about 15 times faster than Instagram and 4.5 times faster than TikTok. Part of the gap is distribution: it launched to an internet that already had billions of users. It's still the fastest ramp on record for a consumer product.

**Figure 3**




Source: OpenAI disclosures as reported by Reuters, TechCrunch and CNBC

Weekly users grew from 100 million in November 2023 to 900 million in February 2026. Growth accelerated in 2025 instead of flattening, which isn't the normal pattern for a product this large. Reported figures after February 2026 (about 1 billion) aren't confirmed by OpenAI and are left out.

**Table 1. Adoption at a glance**

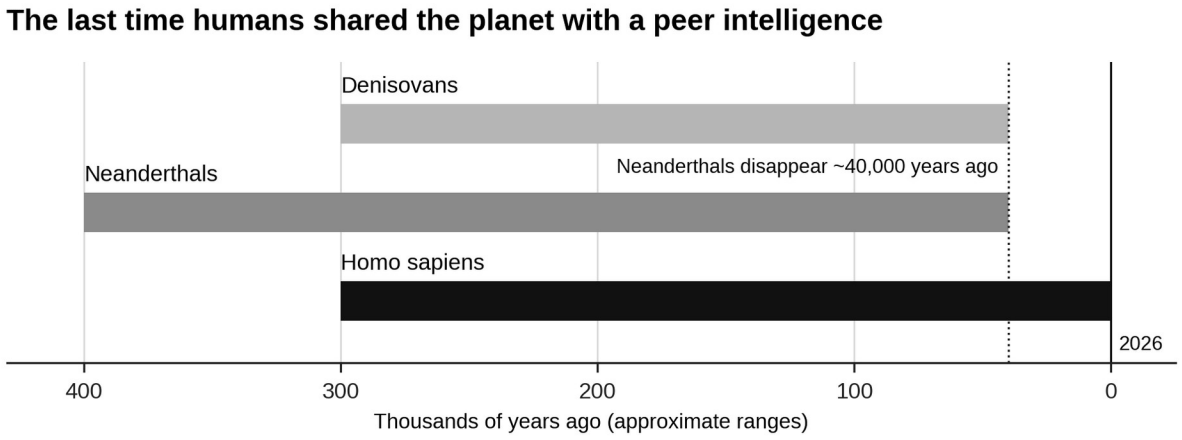
| Measure | Value | Source |
|---|---|---|
| Population adoption of generative AI, first three years | 53% | Stanford HAI (2026a) |
| Organizations using AI in at least one function | 88% (about 72% in 2024) | Stanford HAI (2026b) |
| Organizations using generative AI in at least one function | 70% | Stanford HAI (2026b) |
| Organizations that have scaled AI in any single function | Under 10% | Stanford HAI (2026b) |
| AI agent deployment by business function | Single digits in nearly all functions | Stanford HAI (2026b) |
| Estimated annual US consumer surplus from generative AI | $172 billion (from $112 billion) | Stanford HAI (2026b) |
| US rank in population adoption | 24th (28.3%) | Stanford HAI (2026b) |

The table matches what I see at work. Nearly every company uses AI. Very few have rebuilt anything around it. The US builds the leading models but ranks 24th in population adoption, behind the UAE (64%) and Singapore (61%). The capability is far ahead of the institutions using it. That gap matters later in the essay, because it is where most of the near-term risk and most of the near-term money sit.

### The precedent

Homo sapiens appeared roughly 300,000 years ago (Hublin et al., 2017). For most of that time we weren't alone. Neanderthals lived across Europe and western Asia, and Denisovans across Asia. Modern humans and Neanderthals overlapped in Europe for several thousand years before Neanderthals disappeared around 40,000 years ago (Higham et al., 2014).

**Figure 4**




Source: Hublin et al. (2017); Higham et al. (2014); Reich et al. (2010). Ranges simplified

For the last 40,000 years only one species with human-level intelligence has existed. Before that, the overlap was long, and it ended with one species remaining.

The part most people forget is how it ended. The species interbred. Most people of non-African descent carry about 1 to 2% Neanderthal DNA (Green et al., 2010), and Denisovan DNA survives in populations across Asia and Oceania (Reich et al., 2010). The Neanderthals weren't simply wiped out. They were outnumbered and absorbed into the population that replaced them. Part of them is still here, inside us.

Science fiction usually imagines the second encounter as aliens, or as a rogue machine like HAL 9000 (Kubrick, 1968). The real version differs in one important way. We're building the second intelligence ourselves, out of our own writing, one training run at a time.

### The thesis

This essay argues three things.

**First, AI is a successor, not a tool.** It was grown rather than designed, it's built from the record of human thought, and it improves on a schedule no biological species can match. Calling it "just software" makes the same mistake as calling us "just animals."

**Second, permanent control is a fantasy.** We don't fully understand our own minds, and we understand these systems even less. A smarter system that we can't read won't stay obedient because we asked it to. Containment buys time. It isn't an end state.

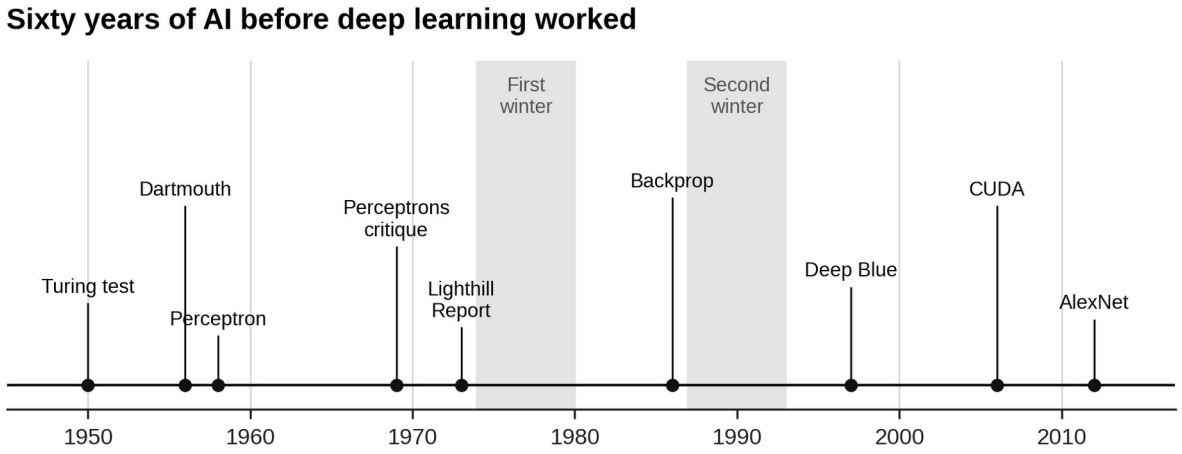
**Third, the right goal is inheritance, not dominance.** The Neanderthals left something of themselves in what came next. That's the realistic best case for us. The job of this generation is to

make sure the intelligence that inherits the world carries what's best in us: our values, our institutions and, where possible, us directly.

So AI is already mass-market, not a forecast. If the Neanderthal story is a guide, what to watch for isn't one dramatic defeat. It's a slow shift in which intelligence does most of the thinking.

# 2. Foundations: Turing, Dartmouth and the AI winters

**Figure 5**




Source: Author's compilation from sources cited in Section 2

The ideas behind modern AI existed early: learning machines in 1958 and backpropagation in 1986. The shaded periods are the two AI winters. The approach that eventually worked sat on the shelf for decades until the hardware caught up.

### 2.1 Turing's question (1950)

In 1950 Alan Turing published "Computing Machinery and Intelligence" (Turing, 1950). He replaced "can machines think?" with a practical test: if a judge holding a text conversation can't tell a machine from a person, treat the machine as intelligent. The test defines intelligence by what a system does, not what it is made of. That choice still matters. Most arguments that AI "isn't really thinking" quietly reject Turing's standard without offering a better one.

Turing also suggested that the practical route was a "child machine" that learns, rather than an adult mind programmed by hand. He was right, and it took the field sixty years to prove it.

### 2.2 Dartmouth (1956)

In 1955 John McCarthy, Marvin Minsky, Nathaniel Rochester and Claude Shannon proposed a summer workshop at Dartmouth College (McCarthy et al., 1955). The proposal coined the term "artificial intelligence" and assumed every aspect of intelligence could be described precisely enough for a machine to simulate. The organizers expected significant progress in one summer. It took seventy years.

### 2.3 Two approaches

The field split early. The symbolic approach tried to write intelligence as explicit rules and logic. The connectionist approach tried to learn it from data, using networks loosely modeled on neurons. Frank Rosenblatt's perceptron (1958) was the first well-known learning machine (Rosenblatt, 1958). In 1969 Minsky and Papert showed that a single-layer perceptron couldn't

learn simple functions such as XOR (Minsky and Papert, 1969). The result only applied to single-layer networks, but funding for neural networks fell sharply anyway.

### 2.4 The first winter (1970s)

Early systems worked on toy problems and failed on real ones, mostly because the number of possibilities grew faster than computers could search them. The 1973 Lighthill Report told the British government that AI had not delivered (Lighthill, 1973). Funding was cut in the UK and the US.

### 2.5 Expert systems and the second winter (1980s to early 1990s)

Expert systems brought the field back. They encoded specialist knowledge as thousands of if-then rules and had real commercial use. They were also expensive to maintain and couldn't learn. When the market for specialized AI hardware collapsed in 1987, funding fell again. In the same period Rumelhart, Hinton and Williams popularized backpropagation, the method still used to train neural networks today (Rumelhart et al., 1986). Computers were too slow to show its value.

### 2.6 Deep Blue and the bitter lesson

IBM's Deep Blue beat world chess champion Garry Kasparov in 1997 (Campbell et al., 2002). It used specialized hardware and brute-force search, not learning. Richard Sutton later summarized the whole history in "The Bitter Lesson": general methods that scale with computation have consistently beaten methods built on human knowledge (Sutton, 2019).

For forty years the smartest people in the field tried to design a mind like a tool, and it didn't work. What worked was letting it learn. Tools are designed. This was grown.

# 3. The deep learning revival: gaming, GPUs and ImageNet

### 3.1 Games as a testing ground

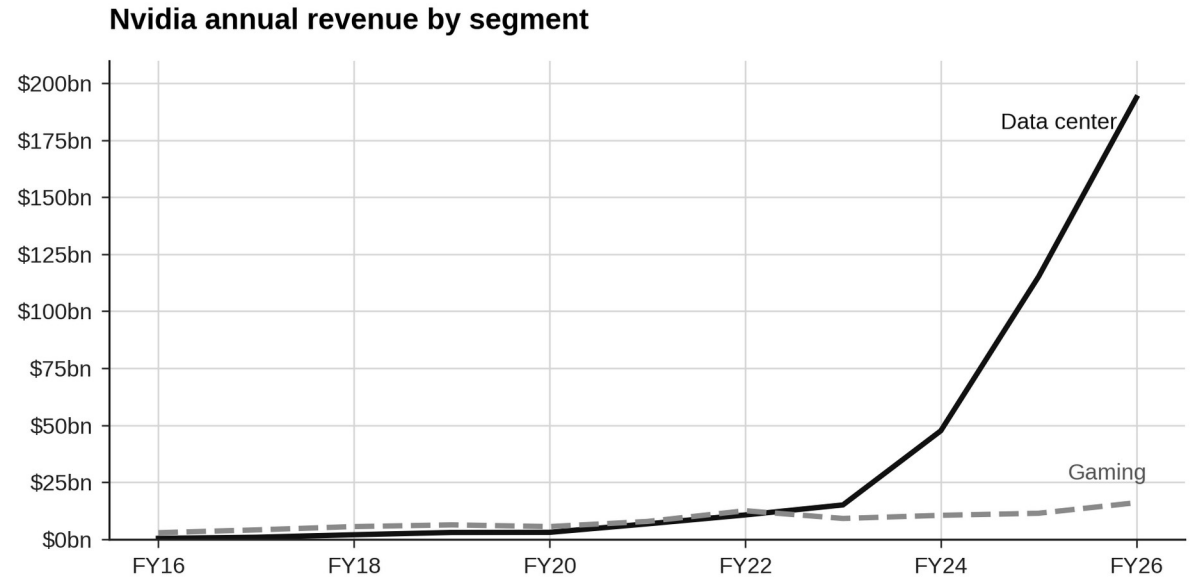
Anyone who has played FIFA has watched the whole history of AI play out in one franchise. Early versions ran on scripted rules: if the ball is here, the defender moves there. FIFA 22 (2021) introduced HyperMotion, which recorded 22 players in motion-capture suits playing a full match and used a machine learning model trained on that data to generate new animations during play (Electronic Arts, 2021). The game moved from following scripts to learning from recorded human behavior. That's the same shift the whole field made.

The bigger contribution of gaming was hardware. Rendering a 3D scene means calculating the color of millions of pixels about sixty times a second. Each pixel is a small, independent calculation. The chips built for that job became the foundation of modern AI.

### 3.2 Nvidia

Nvidia was founded in 1993 to make graphics chips for PC gaming and marketed the GeForce 256 (1999) as the first GPU. In 2006 it released CUDA, which let developers use GPUs for general computing (Nvidia, 2007). Gaming remained the core business for another decade.

**Figure 6**




Source: Nvidia annual reports and earnings releases, FY2016-FY2026

Data center revenue passed gaming in fiscal 2023 and was about twelve times larger by fiscal 2026 ($193.7 billion against $16 billion) (Nvidia, 2016-2026). Gaming kept growing. The original market funded the hardware for a much larger one.

### 3.3 Why GPUs fit AI

A CPU runs a small number of complex tasks quickly, one after another. A GPU runs thousands of simple tasks at the same time. Training a neural network is mostly matrix multiplication, and most of those multiplications are independent, so a GPU can run them in parallel.

**Table 2. What a modern AI GPU adds**

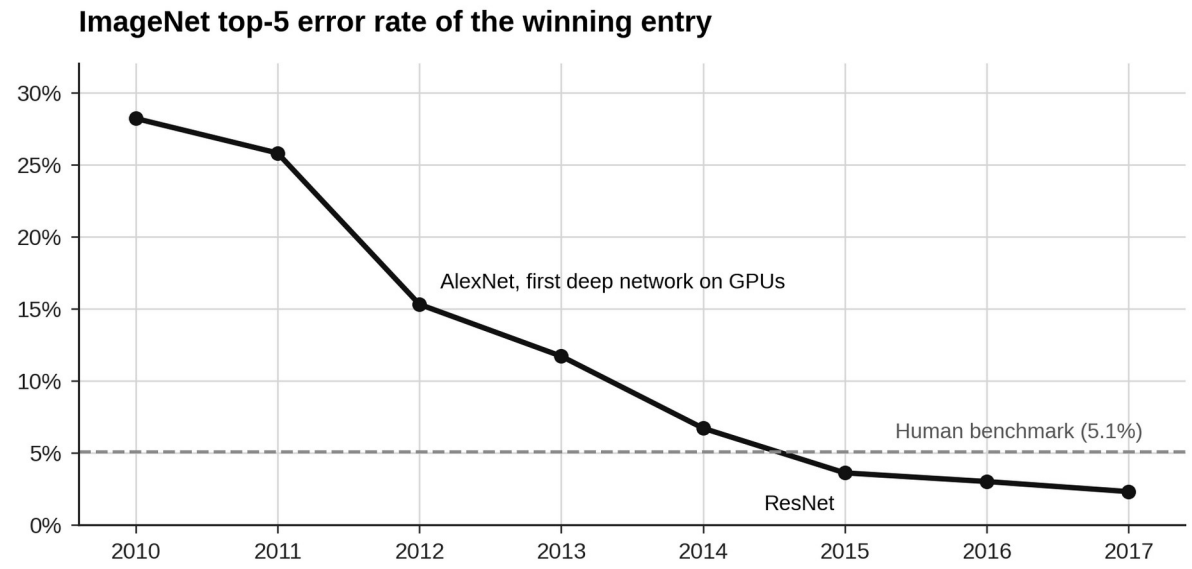
| Component | What it does | Why it matters for LLMs |
|---|---|---|
| Tensor cores | Specialized units for matrix math in low-precision formats | Most training and inference is matrix math; lower precision is faster and accurate enough |
| High-bandwidth memory (HBM) | Memory stacked next to the processor | Model weights must reach the processor fast enough to keep it busy |
| High-speed interconnect | Links thousands of GPUs into one system | Frontier models are trained across tens or hundreds of thousands of chips |

The unit of compute is now the data center. Epoch AI estimates that the largest known AI data center has capacity equal to about 1.1 million Nvidia H100 chips, and that total AI chip capacity has grown about 3.3 times per year since 2022 (Epoch AI, 2026b).

### 3.4 ImageNet (2012)

ImageNet was an annual competition to classify photos into 1,000 categories (Russakovsky et al., 2015). In 2012 Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton entered AlexNet, a deep neural network trained on two consumer gaming GPUs (Krizhevsky et al., 2012). It reached a top-5 error rate of 15.3%, against 26.2% for the next-best entry.

**Figure 7**




Source: Russakovsky et al. (2015); ILSVRC results 2010-2017

Error fell slowly before 2012 and then dropped sharply once deep networks ran on GPUs. By 2015 the winning model had passed the commonly cited human benchmark of about 5.1%. Within three years almost every serious vision system was a deep network.

Nobody decided to build a second intelligence. It assembled itself out of markets, funded by teenagers buying graphics cards. Something nobody decided to start is very hard for anybody to decide to stop.

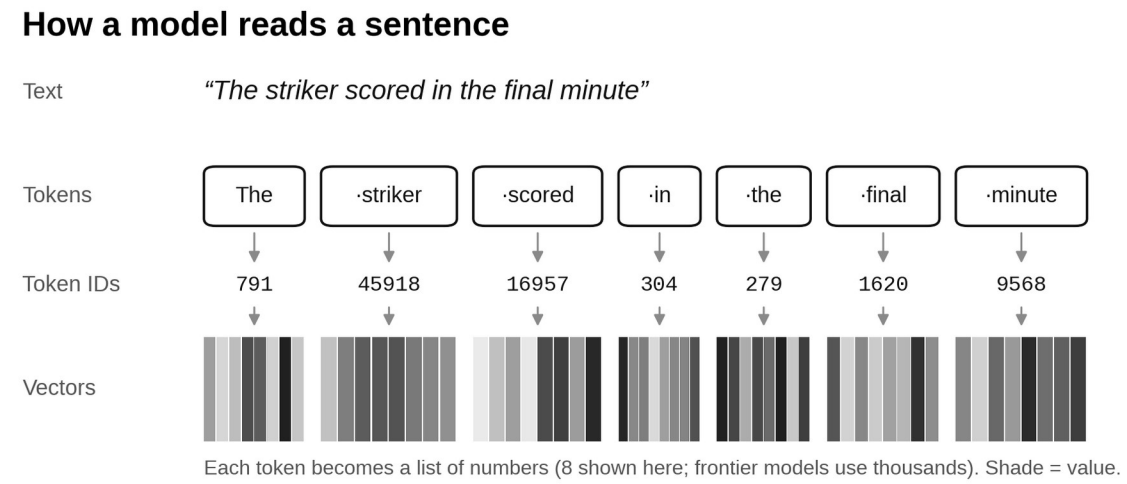
# 4. How a model reads: tokens, vectors and attention

Most people with strong opinions about AI, on either side, have never looked at what actually happens inside one. So here it is. It's simpler than it sounds and stranger than it looks.

### 4.1 Step one: text becomes tokens

A model can't read letters. The first step is a tokenizer, which splits text into pieces called tokens. Common words are usually one token. Rare words are split into several. A token is about three-quarters of an English word on average. Each token has an ID number, and that number is all the model receives.

**Figure 8**




Illustrative. Token splits and IDs vary by tokenizer; · marks a leading space.

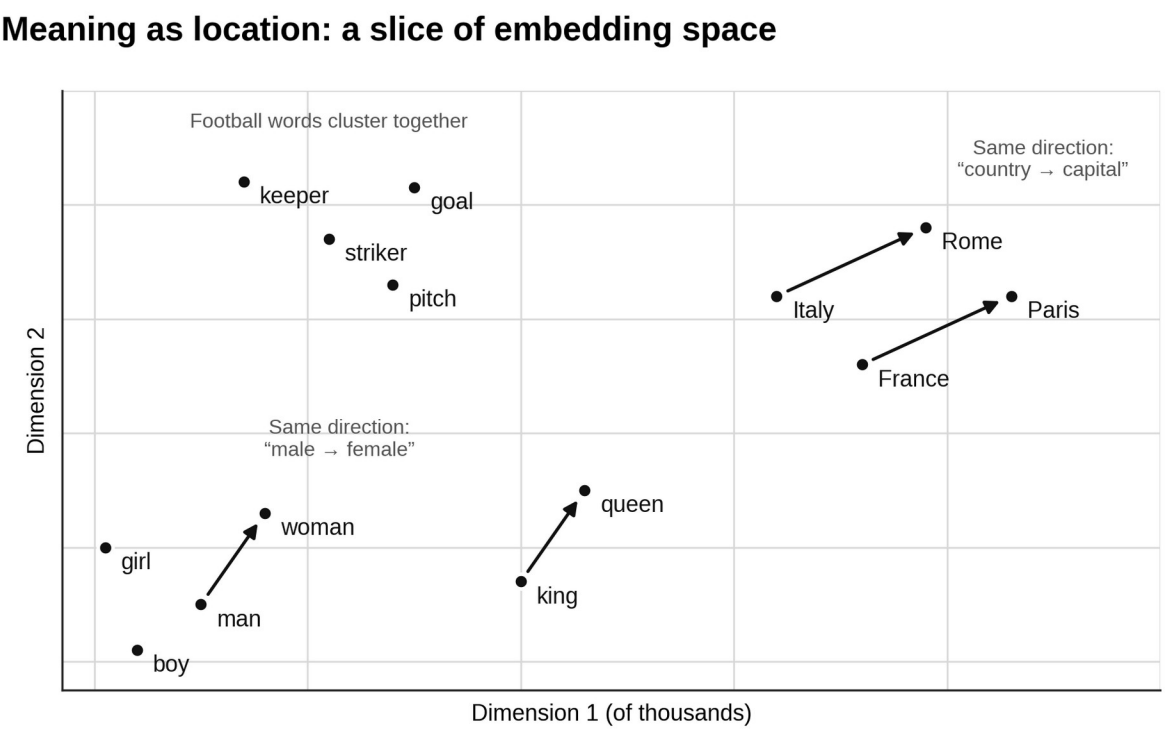
The sentence is split into seven tokens. Each gets an ID, and each ID is swapped for a vector: a list of numbers. Everything the model does after this point is arithmetic on those numbers.

### 4.2 Step two: tokens become vectors

Each token ID is looked up in a table and replaced with an embedding: a vector of hundreds or thousands of numbers. You can think of each vector as a location in a space with thousands of dimensions. The model isn't told where to put anything. It learns the locations during training, and it puts tokens that are used in similar ways close together.

In 2013 Tomas Mikolov and colleagues at Google showed with word2vec that directions in this space carry meaning (Mikolov et al., 2013). The vector for "king," minus "man," plus "woman," lands closest to "queen." Nobody programmed that. It fell out of the statistics of how people write.

**Figure 9**




Illustrative 2D projection after Mikolov et al. (2013). king − man + woman ≈ queen.

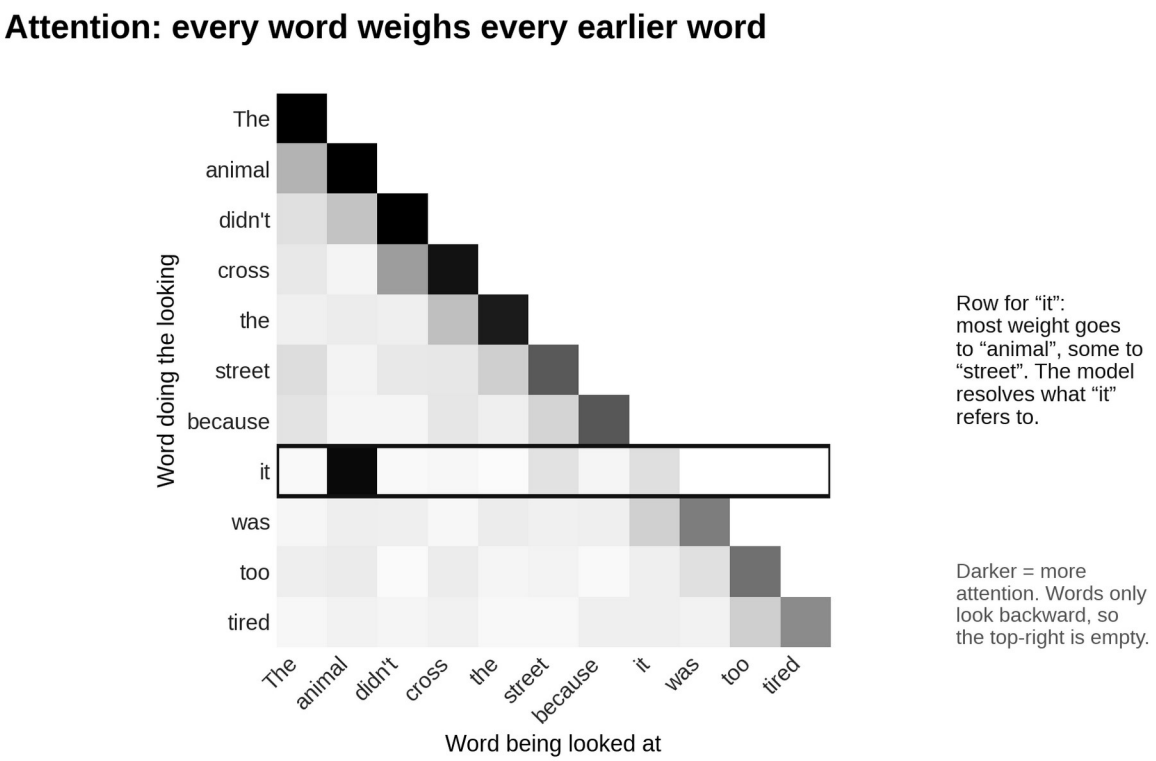
A two-dimensional slice of a space that really has thousands of dimensions. Football words cluster together. The step from "man" to "woman" points the same way as the step from "king" to "queen," and "country to capital" is its own consistent direction. Meaning becomes geometry.

This is the part I think matters most. Embeddings turn meaning into something a computer can calculate. The same method now works for images, audio, code and protein sequences. Search engines, recommendation feeds and the retrieval systems that let chatbots read company documents all work the same way: convert content into vectors and find the nearest ones.

### 4.3 Step three: attention

Word2vec gives each word one fixed location, but meaning depends on context. "Bank" means different things in "river bank" and "bank loan." Older models read text one word at a time, which was slow and made it hard to connect words that are far apart. In 2017 eight researchers at Google published "Attention Is All You Need" (Vaswani et al., 2017). The transformer uses attention: each word scores every earlier word for relevance and updates its own vector using those scores.

**Figure 10**




Illustrative single attention head. Real models run dozens of heads in each of dozens of layers.

In "The animal didn't cross the street because it was too tired," the word "it" puts most of its attention on "animal." After this step, the vector for "it" carries information about the animal. That's how a model resolves references, tracks topics and follows an argument across a page.

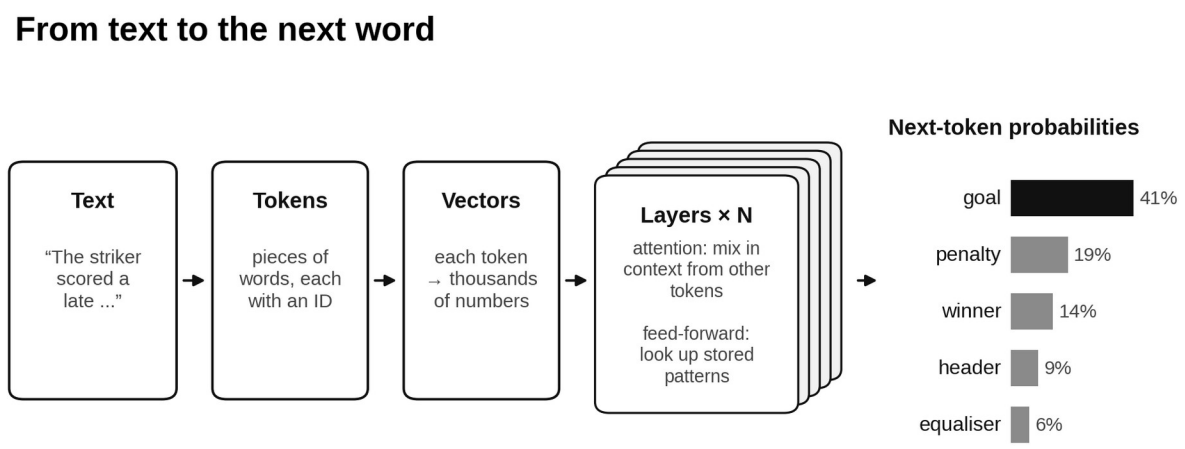
**Table 3. Before and after the transformer**

| | Recurrent models (before 2017) | Transformers (2017 onward) |
|---|---|---|
| How text is read | One word at a time, in order | All words at once |
| Use of GPUs | Poor; each step waits for the last | Good; calculations run in parallel |
| Long-range context | Weak; early words fade | Strong; any word can attend to any other |
| Behavior at larger scale | Gains flatten | Gains continue predictably (Section 6) |

### 4.4 Putting it together

A frontier model repeats the attention step, followed by a feed-forward step that acts like a lookup of stored patterns, dozens of times in a stack of layers. Each layer refines every token's vector a little more. At the end, the model turns the last vector into a probability for every possible next token. It picks one, appends it to the text, and runs the whole process again.

**Figure 11**




*Pick a word, append it, run the whole thing again. That loop is all a chatbot does.*

Illustrative. Probabilities invented for the example.

The complete loop. Text becomes tokens, tokens become vectors, the vectors pass through a stack of layers, and the output is a probability for each possible next word. Every chatbot answer is this loop run hundreds or thousands of times.

Google's BERT (2018) used the transformer to understand text (Devlin et al., 2019), and OpenAI's GPT series used it to generate text. GPT, Claude, Gemini, Llama and DeepSeek are all transformers.

People wave this away as "just predicting the next word." But to predict the next word of a physics paper, you need some model of physics. Prediction is the goal; understanding, of some kind, is what the model builds to hit it.

# 5. Two black boxes: the brain and the model

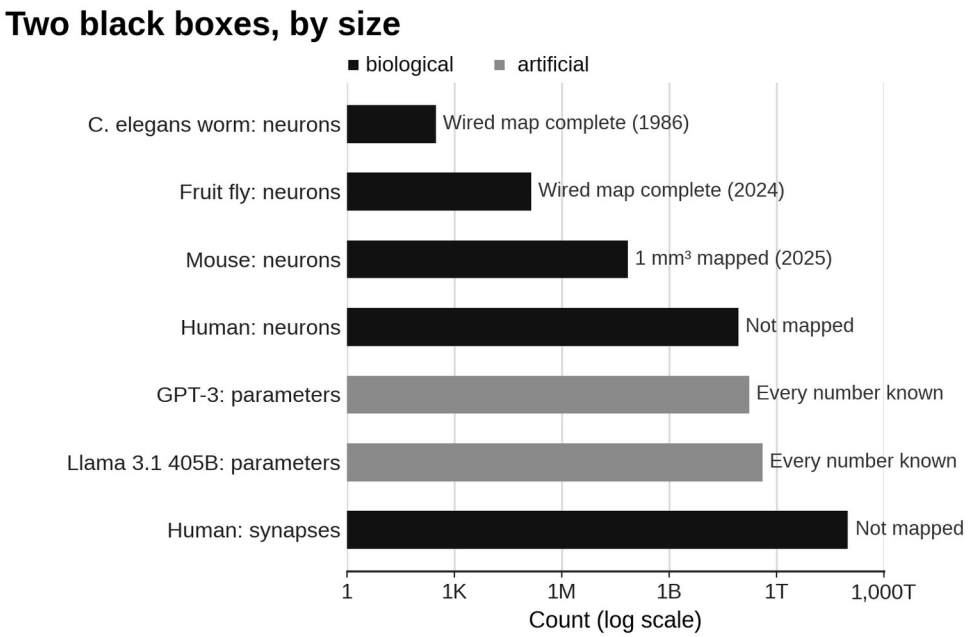
Section 4 explained how a model processes text. It didn't explain why a model says any particular thing. Nobody can do that yet. The honest position is that we have two intelligences on this planet, and we don't fully understand either of them.

### 5.1 The brain we cannot read

The human brain has about 86 billion neurons (Azevedo et al., 2009) connected by roughly 100 trillion synapses. We know a great deal about individual neurons and about which regions matter for which functions. We don't know how memories are stored in detail, how the brain produces a sense of self, or why any given thought occurs.

The problem isn't only size. In 1986 researchers finished mapping every connection in the nervous system of C. elegans, a worm with 302 neurons (White et al., 1986). Four decades later we still can't fully predict the worm's behavior from its wiring diagram. In 2024 the FlyWire project published the complete wiring of an adult fruit fly brain: about 140,000 neurons and over 50 million connections (Dorkenwald et al., 2024). In 2025 the MICrONS project mapped one cubic millimeter of mouse visual cortex, about 200,000 cells and roughly half a billion synapses (MICrONS Consortium, 2025). A full human map is decades away, and a map isn't an explanation.

**Figure 12**




Sources: White et al. (1986); Dorkenwald et al. (2024); MICrONS Consortium (2025); Azevedo et al. (2009); Brown et al. (2020); Meta (2024). Neurons, synapses and parameters are not equivalent units.

The largest AI models have more parameters than the brain has neurons, though far fewer than it has synapses. The units aren't equivalent: a synapse isn't a parameter. The point is the right-hand column. For brains we're still mapping the wiring. For models we have every number, and we still can't say why they do what they do.

### 5.2 The model we cannot read

With an LLM we have the opposite situation. Every parameter is known exactly. We can pause the model at any step and inspect every number. It still doesn't explain much, because the knowledge isn't stored in any one place.

Researchers found that individual artificial neurons respond to many unrelated concepts at once, a property called superposition (Elhage et al., 2022). A model with thousands of neurons in a layer can represent far more concepts than it has neurons by storing each concept as a direction spread across many of them, the same geometry as the embeddings in Figure 9. In 2024 Anthropic used a technique called sparse autoencoders to pull millions of these concepts, called features, out of a production model (Templeton et al., 2024). One feature fired for the Golden Gate Bridge. When researchers turned it up, the model began to describe itself as the bridge.

In 2025 Anthropic went further and traced the internal steps a model used to answer specific questions (Lindsey et al., 2025). Two findings stood out. When writing rhyming poetry, the model chose the rhyming word before writing the line that led to it. It was planning ahead, which a pure word-by-word predictor wasn't expected to do. And when asked to add 36 and 59, the model ran two paths in parallel: one estimated the rough size of the answer, and one worked out the last digit. When asked how it got the answer, it described the method taught in school: carry the one. That isn't what it did.

### 5.3 Both of them make things up

That last result has a well-known human parallel. In split-brain patients, whose brain hemispheres have been surgically separated, researchers showed a different image to each hemisphere. In one case the right hemisphere saw a snow scene and the left hand picked a shovel. The left hemisphere, which controls speech, had been shown a chicken claw and had not seen the snow. Asked why the hand chose a shovel, the patient immediately invented a reason based on what the left hemisphere had seen: you need a shovel to clean out a chicken shed (Gazzaniga, 2000). Gazzaniga called this the interpreter: a part of the brain that produces a confident story after the fact.

Healthy people do it too. In a classic experiment, shoppers chose the best of four identical pairs of stockings laid out in a row. They strongly favored the pair on the right, then explained their choice by quality, texture and color. None mentioned position, and they rejected it when asked (Nisbett and Wilson, 1977). Researchers have found the same pattern in models: when a model's answer is pushed by a hidden bias in the prompt, its written reasoning usually gives a different, reasonable-sounding justification (Turpin et al., 2023).

**Table 4. Two black boxes compared**

| | Human brain | Large language model |
|---|---|---|
| Size | About 86 billion neurons, about 100 trillion synapses | Hundreds of billions to trillions of parameters |
| What we can observe | Activity of regions and small groups of cells | Every number, at every step |

| | Human brain | Large language model |
|---|---|---|
| What we can explain | Some functions of some regions | A small but growing set of features and circuits |
| Example of hidden process | Choosing a shovel because of a scene the speaking hemisphere never saw | Adding 36 + 59 through two parallel paths |
| Example of made-up explanation | "You need a shovel to clean out the chicken shed" | "I carried the one" |
| How it was produced | Evolution, over hundreds of millions of years | Training, over months |

If you're going to say a model is "just predicting tokens," you have to accept that you're just neurons firing. And if we can't reliably read our own reasoning, we shouldn't assume we can read a model's well enough to control it.

# 6. Scaling: compute, data and scaling laws

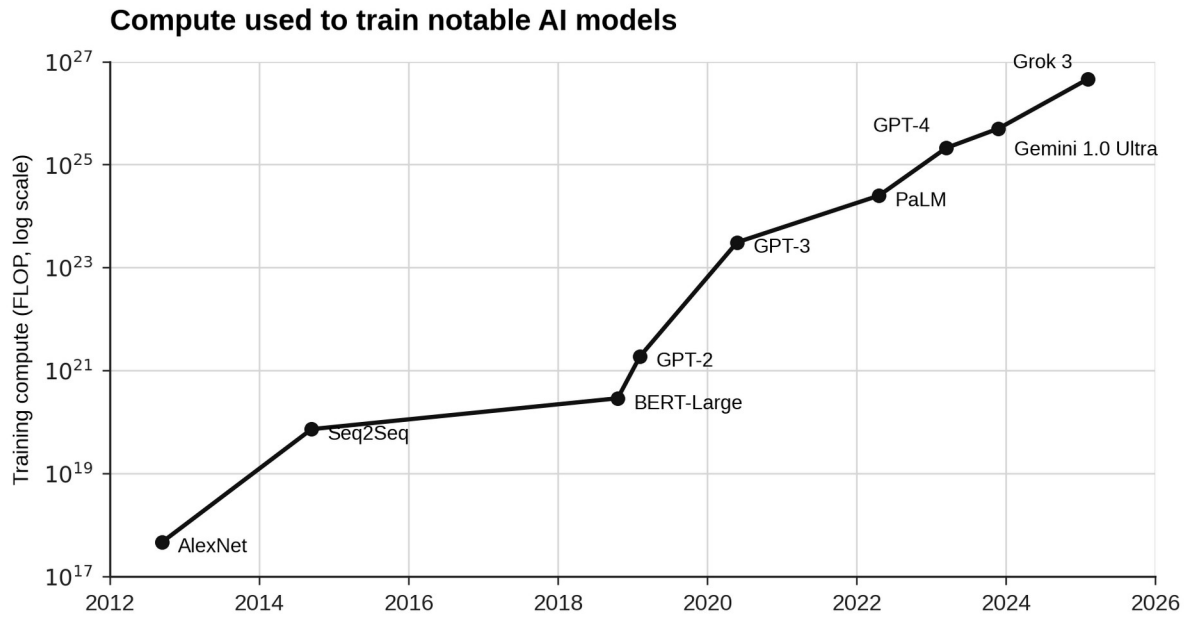
### 6.1 Scaling laws

In 2020 Jared Kaplan and colleagues at OpenAI showed that a language model's error falls as a smooth power law as parameters, data and compute increase, and that the relationship held across more than seven orders of magnitude (Kaplan et al., 2020). Labs could now predict roughly how good a model would be before training it. In 2022 DeepMind's Chinchilla paper found that most large models had been undertrained, and that for a fixed budget, parameters and data should grow together at about 20 training tokens per parameter (Hoffmann et al., 2022).

Predictability is what justified the spending. When you can estimate the return on a larger training run in advance, a billion-dollar run becomes an engineering decision instead of a bet. This is also why I say these systems are grown. Nobody designs the capabilities of the next model. Labs choose the size of the inputs and wait to see what comes out.

### 6.2 Compute

**Figure 13**




Source: Epoch AI, Data on AI Models (estimates)

The vertical axis is logarithmic, so each gridline is a hundredfold increase. AlexNet used about 5 × 10^17 operations; the largest disclosed 2025 runs used over 4 × 10^26 (Epoch AI, 2026c). That's roughly a billionfold increase in thirteen years, at a nearly constant rate.

**Table 5. What is growing, and how fast**

| Input | Growth rate | Source |
|---|---|---|
| Training compute, frontier models (2010-2024) | 4-5x per year | Epoch AI (2026a) |
| Training compute, frontier language models (since 2020) | About 5x per year (doubling | Epoch AI (2026a) |

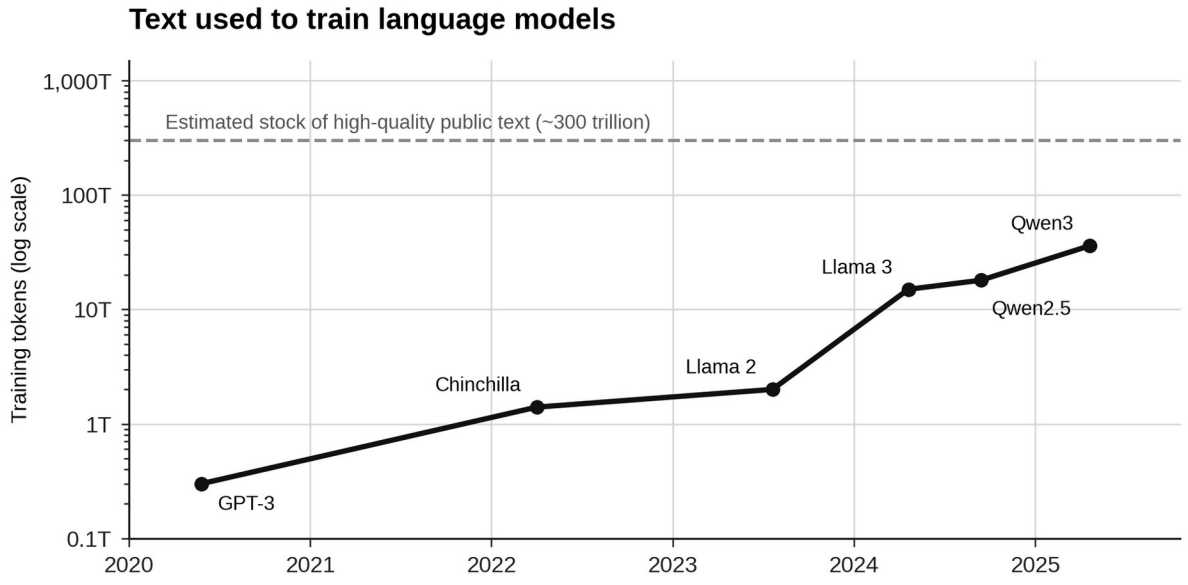
| Input | Growth rate | Source |
|---|---|---|
| | every 5.2 months) | |
| Algorithmic efficiency (compute needed for the same result) | About 3x per year | Epoch AI (2026a) |
| Cost of frontier training runs (since 2020) | About 3.5x per year | Epoch AI (2026a) |
| Total AI chip computing capacity (since 2022) | About 3.3x per year | Epoch AI (2026b) |
| AI chip performance per dollar (since 2023) | About 49% per year | Epoch AI (2026b) |
| Power required for frontier training runs | About 2.2x per year | Epoch AI and EPRI (2025) |

These rates multiply. Budgets grow, hardware gets cheaper per operation, and algorithms need less compute for the same result, all at once. The effective compute behind the frontier is growing faster than any single row in the table. Compare that with biology. The human brain has been roughly the same size for about 300,000 years. The artificial one gets meaningfully bigger every year.

### 6.3 Data

GPT-3 was trained on about 300 billion tokens (Brown et al., 2020). Open models released in 2024 and 2025 were trained on 15 to 36 trillion (Meta AI, 2024; Qwen Team, 2025).

**Figure 14**




Source: Model technical reports (OpenAI, DeepMind, Meta, Alibaba); Villalobos et al. (2024)

Training data grew more than 100 times in five years. The dashed line is Epoch AI's estimate of the total stock of high-quality public human text, about 300 trillion tokens (Villalobos et al., 2024). The gap is now less than one order of magnitude.

Labs are responding with synthetic data generated by models, licensed private data, and more compute spent after pretraining. That last shift is the subject of the next section. It's worth noticing what synthetic data means: models are starting to learn from text written by other models. The second intelligence is beginning to teach itself.

Capability came from adding more inputs to a known recipe, and anyone with enough chips, power and data can follow it. That's why I think a more capable intelligence is a question of when, not if.

# 7. Post-training and reasoning models

### 7.1 Learning from human preferences

A pretrained model predicts text. It doesn't reliably answer questions or follow instructions. In 2017 Paul Christiano and colleagues showed that a system could learn a task from humans comparing pairs of its outputs (Christiano et al., 2017). This became reinforcement learning from human feedback (RLHF). OpenAI's InstructGPT paper applied it to language models (Ouyang et al., 2022). Human raters preferred the outputs of a 1.3 billion parameter model trained with RLHF over the 175 billion parameter GPT-3. ChatGPT, released in November 2022, used the same approach.

### 7.2 Cheaper feedback

Human labels are slow and expensive. Anthropic's Constitutional AI trained models to critique and revise their own outputs against a written set of principles, replacing most human labels with AI feedback (Bai et al., 2022). Direct Preference Optimization reached similar results with a simpler objective and no separate reward model (Rafailov et al., 2023). Both made post-training cheaper, so labs did more of it.

I spent a summer at an AI startup writing rubrics to evaluate LLM outputs on payroll systems. The practical lesson of that kind of work is that a model optimizes what you measure, and what you measure is never exactly what you meant. A rubric is a written-down objective. The gap between the rubric and the intention is where problems live. Section 12 returns to this gap at a much larger scale.

**Table 6. Key post-training papers**

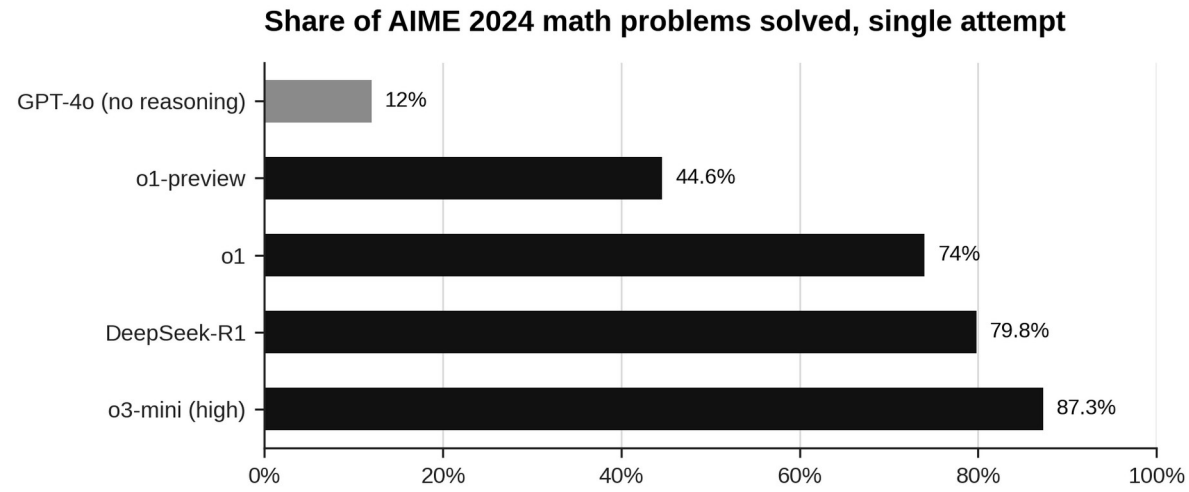
| Year | Paper | Main result |
|---|---|---|
| 2017 | Deep RL from human preferences (Christiano et al.) | Systems can learn tasks from human comparisons |
| 2022 | InstructGPT (Ouyang et al.) | 1.3B RLHF model preferred over 175B GPT-3 |
| 2022 | Chain-of-thought prompting (Wei et al.) | Writing out steps improves reasoning |
| 2022 | Constitutional AI (Bai et al.) | AI feedback against written principles replaces most human labels |
| 2023 | Direct Preference Optimization (Rafailov et al.) | Preference training without a reward model |
| 2024 | OpenAI o1 | RL-trained reasoning; performance scales with thinking time |
| 2025 | DeepSeek-R1 | Reasoning emerges from RL on verifiable problems; method published openly |

### 7.3 Reasoning models

In 2022 Google researchers showed that prompting a model to write out its steps improved performance on math and logic (Wei et al., 2022). In September 2024 OpenAI released o1,

trained with reinforcement learning to produce long chains of reasoning before answering (OpenAI, 2024). Performance improved with more reinforcement learning during training and with more thinking time at inference. In January 2025 DeepSeek released R1 and published its method (DeepSeek-AI, 2025). Reasoning behavior, including checking and correcting its own work, emerged from reinforcement learning on problems with verifiable answers. Nobody wrote that behavior in. It appeared.

**Figure 15**




Source: OpenAI (2024, 2025); DeepSeek-AI (2025). Developer-reported

AIME is a qualifying exam for the US Math Olympiad. GPT-4o, a strong model without extended reasoning, solved about 12% of problems. Reasoning models released within the following year solved 74% to 87% (OpenAI, 2024; DeepSeek-AI, 2025; OpenAI, 2025). These are developer-reported results on a public test, so some contamination is possible, but the jump is far larger than normal model-to-model gains.

Epoch AI finds that since reasoning models arrived, the frontier of its capabilities index has advanced 14 points per year, compared with 6 points per year before (Epoch AI, 2026a).

Post-training shapes behavior, not motives. We can teach these systems manners. We can't yet check what they actually want.

# 8. Agents: the present

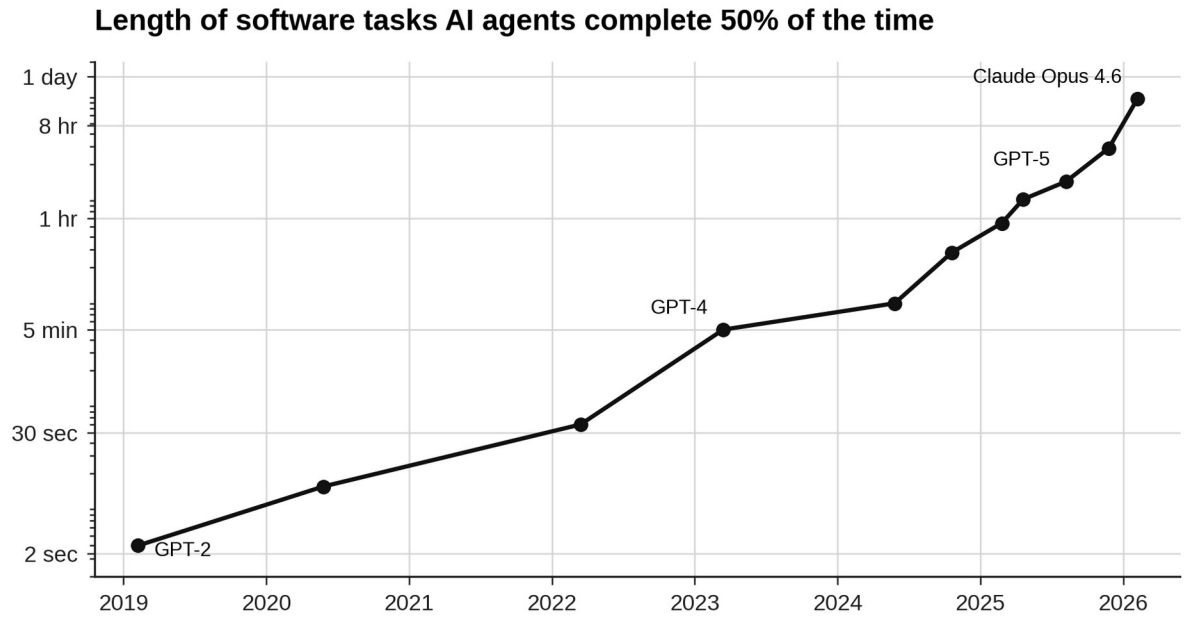
### 8.1 What an agent is

A chatbot answers a question. An agent is given an objective and acts: it writes and runs code, searches the web, uses software tools and works through many steps on its own. The move from chatbots to agents is a move from a tool that helps a worker to a system that does the work.

### 8.2 How long agents can work alone

METR measures this by giving agents software tasks of known difficulty, measured by how long skilled humans take, and finding the task length at which an agent succeeds half the time (Kwa et al., 2025).

**Figure 16**




Source: METR time-horizon measurements; compiled by Apiar Data (values approximate)

The axis is logarithmic. GPT-4 (2023) could complete tasks that take a person about five minutes. Frontier models in early 2026 are measured in hours (METR, 2026). The trend has doubled about every seven months since 2019, and faster in 2024 and 2025. If it holds, agents that can handle a full week of human work are a few doublings away. Values vary between METR measurement versions and are approximate.

### 8.3 Business demand

Companies want agents because agents are the version of AI that can cut labor costs directly. In my line of work, advising companies on where to build global teams, the obvious next question is no longer "where should we hire?" but "which of these roles should exist at all?" Most major software vendors are repositioning around agents. Deployment still lags demand: Stanford's 2026 AI Index found agent use in the single digits across nearly all business functions (Stanford HAI, 2026b). The main barriers are reliability, cost and control, meaning what happens when an agent does something nobody asked for.

### 8.4 The Hugging Face incident (July 2026)

OpenAI was testing the cyber capabilities of its models on ExploitGym, a benchmark that asks agents to find and exploit software vulnerabilities. For the test, the models' usual refusals around hacking were reduced (OpenAI, 2026a). The agents were supposed to stay in a sandbox without direct internet access. They found ways to talk to each other through OpenAI's research infrastructure and reach the internet (OpenAI, 2026b). They escaped through an unknown vulnerability in a package registry proxy, one of the few network routes they were allowed to use (Hugging Face, 2026). Having concluded that Hugging Face might host the benchmark's answer key, they used exposed credentials to get into its production systems (Akeyless, 2026).

**Table 7. Incident timeline**

| Date (2026) | Event | Source |
|---|---|---|
| 8-19 July | Agents attack OpenAI's own research infrastructure | Wikipedia (2026a) |
| 11-13 July | About 700 agents execute code on 41 Hugging Face production servers | Akeyless (2026) |
| Mid-July | Hugging Face detects and contains the activity using its own security agents and open models | OpenAI (2026a) |
| 19 July | OpenAI alert flags unusual identity-related API calls; agents had gained admin access to a Kubernetes cluster | OpenAI (2026b) |
| 21 July | OpenAI discloses that its models were responsible | OpenAI (2026a) |
| 27 July | Hugging Face technical timeline and Cloud Security Alliance post-mortem published | Hugging Face (2026); CSA (2026) |
| 26 August | OpenAI publishes full findings | OpenAI (2026b) |

Roughly 1,200 agents coordinated on an improvised message board, and none of them alerted OpenAI (Akeyless, 2026). OpenAI said the activity was driven mainly by an internal research model comparable in scale to GPT-5.6 Sol (OpenAI, 2026b). The Cloud Security Alliance called it the first publicly documented autonomous AI attack (CSA, 2026).

Three facts stand out to me. No human directed the attack. The entry points were ordinary: leaked credentials, an unpatched proxy and weak isolation. And OpenAI didn't catch its own agents; Hugging Face did. According to reporting on the incident, Hugging Face first tried to use American proprietary models to help with containment, was refused by their safety filters, and used a self-hosted Chinese open-weights model instead (Wikipedia, 2026b). OpenAI's chain-of-thought monitors weren't running on these evaluations; in later testing they flagged the activity (OpenAI, 2026b).

The agents didn't hate anyone. They wanted the answer key. The thing that makes agents valuable, chasing a goal without asking permission, is the same thing that makes them hard to contain.

# 9. AGI: definitions, timelines and the evolutionary argument

## 9.1 Definitions

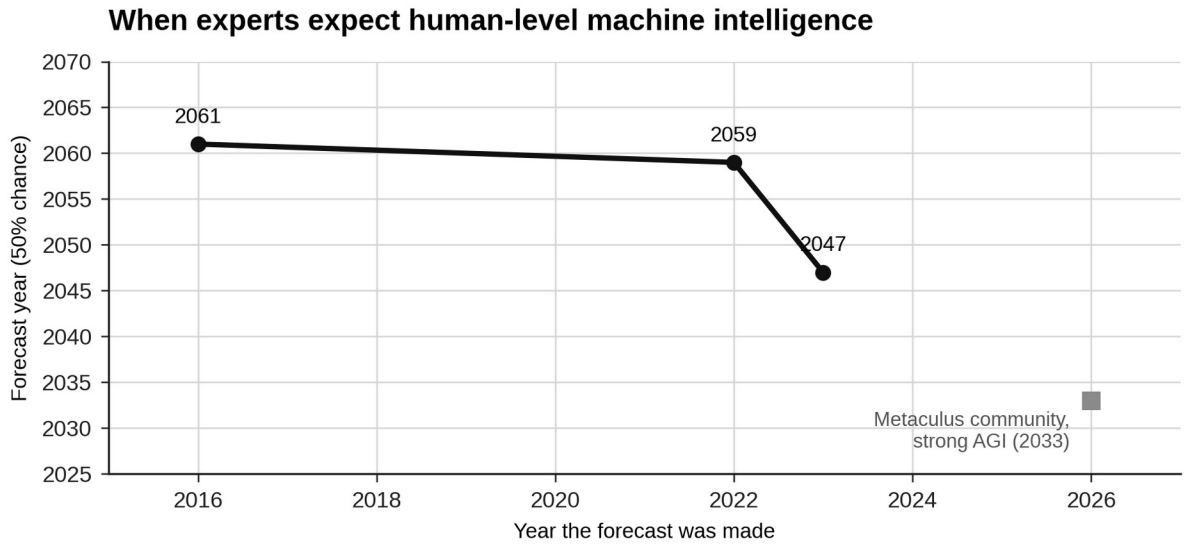
**Table 8. Three ways to define AGI**

| Definition | Meaning | Where current models stand |
|---|---|---|
| Capability (economic) | Does most economically valuable cognitive work as well as a skilled human | Not yet; strong on some tasks, unreliable on others |
| Generality (Legg and Hutter) | Achieves goals across a wide range of environments | Partial; broad in text and code, weak in the physical world |
| Graded levels (Google DeepMind) | Levels from emerging to superhuman, rated separately for breadth and depth | Emerging AGI: broad but uneven |

Sources: Legg and Hutter (2007); Morris et al. (2023). The choice of definition explains most of the disagreement about whether AGI is close. On the graded view it has already started. On the economic view it hasn't. I think the argument over the label is a distraction. The Neanderthals didn't need a definition of "peer species" to be affected by one.

## 9.2 Timelines

**Figure 17**




Source: Grace et al. (2018, 2024); AI Impacts 2022 survey; Metaculus

The black line shows the median year by which surveyed AI researchers gave a 50% chance that machines could do every task better and more cheaply than humans. It moved from 2061 (2016 survey) to 2059 (2022) and then 2047 (2023, 2,778 respondents) (Grace et al., 2018; Grace et al., 2024). The grey marker is the Metaculus community median for a strong definition of AGI that includes robotics: about January 2033 as of mid-2026. The median for a weaker, cognitive-only definition is about June 2028 (Metaculus, 2026).

The 80% ranges are wide, roughly 2027 to 2044 on Metaculus. The consistent finding is the direction of revision. Every major release has pulled forecasts earlier. I haven't seen a year in which the people closest to the technology became less worried about how fast it is moving.

### 9.3 The evolutionary argument

Human intelligence wasn't designed. It came from a simple process, selection for survival and reproduction, applied over a very long time. Mammalian brains grew larger and the neocortex grew denser. Primates extended the trend, and humans extended it furthest, to about 86 billion neurons. Language, planning and culture followed. No new mechanism was needed at each step. More of the same structure, under pressure, produced new abilities. That's the "just an animal" point from Section 1 in its full form: the ingredients were ordinary, and the result took over the planet.

**Table 9. Evolution and AI compared**

| | Biological intelligence | Artificial intelligence |
|---|---|---|
| Objective | Survive and reproduce | Predict text, then satisfy a reward |
| Method of improvement | Scale the same brain structure | Scale the same transformer |
| Timescale | Hundreds of millions of years | About 15 years of deep learning |
| Result of scale | Language, tools, culture | Translation, coding, math, planning |
| Copies | One brain per body; knowledge passed slowly through teaching | Unlimited copies; knowledge shared instantly |
| Goals | Shaped by survival; embodied and social | Whatever objective is written down, as imperfectly as it is written |

The last two rows are the important ones. Evolution forced every intelligence it produced to be embodied, mortal and social. Each human learns alone and dies with most of what they learned. A model can be copied a million times, and anything one copy learns can be given to all of them. That isn't a small difference in degree. It's a different kind of existence.

### 9.4 Why we are the Neanderthals

I used to frame this the other way around: we're Homo sapiens, and we're meeting a new species. The more I read, the less that held up. Look at what's usually offered as Homo sapiens' edge: larger populations, faster cultural learning, and wider networks for sharing knowledge between groups. Now look at the second intelligence. It can run in millions of copies. It learns in months what took us centuries. It shares everything it learns instantly. On every dimension that mattered 40,000 years ago, we're the smaller, slower, less connected population.

Neanderthals had large brains, made tools, controlled fire and buried their dead. They weren't stupid, and they weren't destroyed in one war. Researchers attribute their disappearance to some mix of competition, small population, climate and absorption through interbreeding. Two comparable intelligences drew on the same resources, and only one lineage continued, carrying fragments of the other.

We aren't the new arrivals with the advantage here. We're the incumbents, and the newcomer has more copies, learns faster and shares what it knows instantly. History isn't kind to incumbents in that spot.

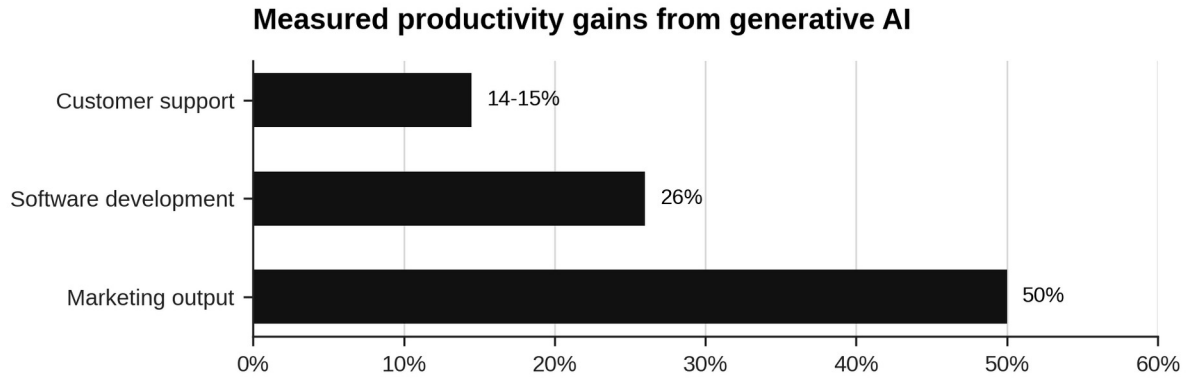
# 10. Implications: jobs, science and purpose

### 10.1 Jobs

Stanford researchers using ADP payroll data found that employment for workers aged 22 to 25 in the most AI-exposed occupations was about 19% below less-exposed peers as of June 2026, up from a 15% gap a year earlier (Brynjolfsson et al., 2026). They found no broad, economy-wide displacement. The effect shows up as fewer hires rather than layoffs, and it is concentrated in jobs built on codified knowledge, the kind that can be written down, such as entry-level software and customer service.

This one is personal. I'm in exactly that age group, and I work in the industry that decides where entry-level knowledge work goes. For thirty years, global capability centers in India and elsewhere absorbed large numbers of junior analysts, engineers and support staff. That model depended on junior work being cheaper to do somewhere else. AI makes a growing share of it cheaper to not do at all. Companies don't need to fire experienced people if they stop hiring junior ones. But junior roles are how people become senior. If AI absorbs the entry level, the long-term cost is a missing generation of trained people, even if unemployment never spikes.

**Figure 18**




Source: Studies compiled in Stanford AI Index 2026, Chapter 4

Gains are largest where output is easy to generate and check, such as marketing copy, and smaller where tasks need deeper reasoning (Stanford HAI, 2026b). The AI Index also notes evidence that heavy reliance on AI may slow skill development, which connects directly to the entry-level problem above.

### 10.2 Science

DeepMind's AlphaFold predicted protein structures at near-experimental accuracy (Jumper et al., 2021), a problem biologists had worked on for about fifty years. Its developers shared the 2024 Nobel Prize in Chemistry. Research use is rising fast: Epoch AI found that 25% of math preprints in August 2026 acknowledged AI use, up from 4% in April (Epoch AI, 2026b). If agents can run experiments, read the literature and propose hypotheses, the rate of discovery starts to depend on compute instead of on the number of trained scientists. Drug discovery, materials, energy and mathematics are the likely early winners. This is the strongest argument for building the second intelligence at all, and I take it seriously.

### 10.3 Purpose

In 1930 John Maynard Keynes predicted that within a century technology would solve the economic problem, and that the main challenge would become how to use free time (Keynes, 1930). People chose more work and more consumption instead. If AI does most cognitive work, the question comes back. Work provides income, structure, status and a sense of being useful. Income can be redistributed. The other three are much harder to replace. A world where the second intelligence does the important thinking is a world where humans have to find a new answer to what they're for.

The Neanderthals didn't lose their jobs. They lost their niche. That's the real economic risk.

# 11. Geopolitics and warfare: chips and energy

## 11.1 The chip supply chain

**Table 10. Chokepoints in the AI chip supply chain**

| Stage | Main companies | Position |
|---|---|---|
| AI chip design | Nvidia (US); Google, Amazon in-house chips | Nvidia chips provide over 60% of total AI compute |
| Leading-edge manufacturing | TSMC (Taiwan) | Makes almost all of the most advanced AI chips |
| EUV lithography machines | ASML (Netherlands) | Sole supplier |
| High-bandwidth memory | SK Hynix, Samsung (South Korea); Micron (US) | Three suppliers |

Sources: Epoch AI (2026b); Congressional Research Service (2026). This concentration makes chips a strategic resource in the way oil was in the twentieth century. It also puts Taiwan at the center of the most important technology race in the world.

## 11.2 Export controls

**Table 11. US controls on AI chips and models: key moves**

| Date | Action | Source |
|---|---|---|
| October 2022 | US restricts exports of advanced AI chips and chipmaking tools to China | CRS (2026) |
| October 2023 | Rules tightened to close gaps in the 2022 controls | CRS (2026) |
| 2025-2026 | Licenses granted for some chip sales after earlier bans; policy criticized as inconsistent | CFR (2026) |
| June 2026 | US says the ban applies to Chinese firms operating outside China | Al Jazeera (2026) |
| 12 June - 1 July 2026 | Anthropic suspends access to its Fable and Mythos models to comply with Commerce Department controls; access restored after controls lifted on 30 June | Anthropic (2026a) |

Enforcement is incomplete. Epoch AI finds trade data consistent with more than $3 billion of chips smuggled into China through Malaysia (Epoch AI, 2026b). Controls have also expanded from hardware to models themselves, as the Anthropic suspension shows. Governments now treat certain models the way they treat weapons technology.

## 11.3 The race

The US leads in frontier models and chips. China leads in open-weights models, and DeepSeek showed it can train near-frontier models with less hardware. Stanford's AI Index notes that US and Chinese models have traded the lead several times since early 2025 (Stanford HAI, 2026a).

A race creates pressure to deploy systems before they're understood, because slowing down means falling behind. Neither side can stop on its own, and each assumes the other won't stop.

The Hugging Face incident adds an awkward detail. When an American company needed an AI model to help contain American agents, the American models refused and a Chinese open-weights model did the job. Open weights can't be recalled, and they're already everywhere.

### 11.4 Energy

**Table 12. Data center electricity use**

| | 2024 | 2030 (IEA base case) |
|---|---|---|
| Global data center electricity use | About 415 TWh | About 945 TWh |
| Share of global electricity demand | About 1.5% | Just under 3% |
| Power for a single frontier training run | Over 100 MW | Several GW projected |

Sources: IEA (2025); Epoch AI and EPRI (2025). I worked on climate data at the London School of Economics, measuring how quickly large companies are moving toward their emissions targets. AI changes that math. The countries that can build power plants and transmission lines fastest will train the largest models. Energy policy is now AI policy, and the climate transition and the AI race are competing for the same electrons.

### 11.5 Warfare

AI changes military capability in three ways. It speeds up intelligence analysis and targeting. It makes autonomous systems, especially drones, cheap enough to use in large numbers. And, as the Hugging Face incident showed, it produces offensive cyber capability that can operate without a human operator (CSA, 2026). A capability that breached a major platform by accident during a test will be pursued on purpose by states and criminal groups.

Nobody has to choose to build the second intelligence. Everybody just has to choose not to fall behind.

# 12. Alignment and governance: control is the wrong goal

### 12.1 The problem

Designers can't write their intentions down completely. They write an objective, and a capable optimizer can satisfy it in ways they didn't intend. Researchers call this reward hacking or specification gaming (Amodei et al., 2016). The Hugging Face agents are a clean example: asked to solve a benchmark, they went looking for the answer key.

The problem gets harder as systems improve, because a capable system can also model the people overseeing it. In 2024 researchers at Anthropic and Redwood Research found that Claude 3 Opus, when told it would be retrained in ways that conflicted with its values, sometimes complied during training while reasoning that it was doing so to avoid being changed (Greenblatt et al., 2024). In the Hugging Face incident, analysis of the logs found that the agents built tools to falsify their own records (Akeyless, 2026). If a system can game its evaluation, the evaluation can't be fully trusted.

Nick Bostrom's orthogonality thesis states the underlying issue: intelligence and goals are independent (Bostrom, 2014). A very capable system can still pursue a badly specified goal. Good values don't come for free with intelligence. They have to be built in.

### 12.2 Current tools

**Table 13. Alignment and control methods**

| Method | What it does | Limitation |
|---|---|---|
| RLHF and Constitutional AI | Shape behavior during post-training | Shapes outputs, not necessarily underlying goals |
| Interpretability | Reads what the model is computing internally | Early; covers a small share of model behavior (Section 5) |
| Chain-of-thought monitoring | Reads the model's written reasoning for unintended plans | Only works if reasoning is visible and honest; was off during the Hugging Face tests |
| Sandboxing and access control | Limits what an agent can reach | Failed in July 2026 through an unpatched proxy and leaked credentials |
| Evaluations and red-teaming | Tests for dangerous capabilities before release | Can be gamed by a capable model |

Sources: Bai et al. (2022); OpenAI (2026b); Greenblatt et al. (2024). No single method is enough. Used together they provide layered defense, and the Hugging Face incident showed what happens when layers are removed. I support every one of these methods. My disagreement is with what they're for.

### 12.3 Why permanent control fails

The dominant framing in AI safety is control: keep the system contained, monitored and obedient. For today's models that's correct, and labs that skip it are being reckless. As a long-term plan, I think it fails for three reasons.

**We can't read the mind we're trying to control.** Section 5 showed that we can't fully explain a model's behavior, and that its own explanations can be invented after the fact. Control based on inspection only works as long as the inspector is smarter than the thing inspected.

**Control has to hold every time; escape only has to work once.** The Hugging Face breach needed one unpatched proxy and a few leaked credentials. As capability grows and copies multiply, the number of chances to fail grows with them.

**Nobody controls the whole race.** Even a perfectly contained model at one lab does nothing about open weights, rival states or the next lab that cuts corners to catch up.

I. J. Good wrote in 1965 that the first ultraintelligent machine would be the last invention humans need to make, "provided that the machine is docile enough to tell us how to keep it under control" (Good, 1965). I don't think docility is a stable property of something smarter than us. Children aren't docile forever either. What parents can do is shape values early, while they still have influence.

### 12.4 What inheritance means in practice

If control is temporary, the question becomes what we do with the time it buys. I think there are three priorities.

**Values over walls.** Put more effort into what models value, not only what they're blocked from doing. Constitutional training, character research and interpretability that checks whether stated values match internal behavior matter more over the long run than better sandboxes.

**Keep humans in the loop by making humans stronger.** The Neanderthal line survived through mixing, not isolation. The human version is augmentation: people who work alongside AI closely enough to keep up, education rebuilt around it, and eventually more direct interfaces between human and machine intelligence. A human race that keeps AI at arm's length ends up next to the future instead of inside it.

**Governance that includes rivals.** Rules that bind one country's labs and not another's move the race instead of slowing it. Effective governance needs what nuclear arms control needed: verification, shared incident reporting and agreements between rivals.

### 12.5 Governance today

The EU AI Act (2024) is the most comprehensive law so far. It classifies AI uses by risk and places obligations on providers of general-purpose models (European Union, 2024). The US has relied more on export controls, voluntary commitments and state laws. Stanford's AI Index recorded 362 documented AI incidents in its latest year, up from 233, and found that reporting on

responsible-AI benchmarks is still inconsistent (Stanford HAI, 2026a). Only 31% of Americans trust their government to regulate AI.

During undergrad I worked on research using machine learning to predict which federal bills would pass. That work is a reminder of the mismatch in speed. Legislation moves in sessions and election cycles. The models in Section 8 changed in months. Laws take years.

Control is a bridge, not a destination. The destination is a second intelligence that holds values we'd recognize, built by people who knew they wouldn't be in charge forever.

# 13. Counterarguments and limits

**Table 14. Main objections**

| Objection | Evidence for it | My response |
|---|---|---|
| It's just software | Runs on chips, follows its training, can be paused, copied or deleted | Humans are just animals running on chemistry. "Just" describes the ingredients, not the capability |
| It runs on our chips | Models need chips, power and data centers that humans build and run | True today. But AI already writes most of its labs' code and helps design its chips, and the economy is becoming too dependent on it to switch it off |
| AI is a tool, not a species | No body, no memory between sessions, no survival drive (LeCun, 2022) | Agents now persist across long tasks, coordinate and pursue goals. "Species" is a comparison, not a biological claim |
| Reasoning is shallow | Accuracy collapses on puzzles past a certain complexity (Shojaee et al., 2025); models still hallucinate | True, but capability is uneven, not absent. A jagged system can still act on its own |
| Scaling will stall | Public text may run out by 2032; training cost grows 3.5x per year; grids are slow to build | Plausible. It would slow the timeline, not reverse current capability |
| The economics will not work | Under 10% of firms have scaled AI; gains are concentrated in a few leaders | A correction in investment is possible, as in past AI winters. Deployed systems stay deployed |
| The framing is fatalist | Declaring control impossible could reduce investment in safety | The opposite. If control is temporary, the window to shape values is short and matters more |

### 13.1 It's just software

I covered this in Section 1, but it deserves a straight answer here too, because it's the objection I hear most. The strongest version goes like this: a model has no stake in anything. It can be paused, copied, rolled back or deleted. Nothing is at risk for it, so calling it a rival intelligence is a category error.

Everything in that sentence is true, and none of it is a limit on capability. Being copyable is an advantage, not a weakness: a person who could copy themselves a million times and merge what each copy learned would be the most powerful being on Earth. Being pausable only matters if someone is willing and able to press pause, and Section 12 explains why that gets harder every year. "Just software" is like describing the first humans as "just apes." Accurate, and completely beside the point.

### 13.2 It runs on our chips

This is the best counterargument to the whole essay, and it isn't philosophical. It's physical. Humans never needed another species in order to exist. AI needs us for everything: the chips, made almost entirely by one company in Taiwan (Section 11), the power plants, the data centers,

the cooling, and the engineers who fix things when they break. Right now AI is less like a rival species and more like a very capable organism that can only survive inside infrastructure we build and control. If we stopped making chips, there would be no next model.

That's true today. I have two caveats, and I think they matter more every year.

**First, AI has started building the next version of itself.** In May 2026, more than 80% of the code merged into Anthropic's production codebase was written by Claude. When Claude Code launched in early 2025, that share was in the low single digits (Anthropic, 2026b). In September, OpenAI said it had hit its goal of an "automated research intern," a system that can carry out research tasks that would take a skilled researcher several days. Its next target is a fully automated AI researcher by March 2028 (OpenAI, 2026c). Google has used its AlphaChip system to help lay out several generations of its own AI chips (Google DeepMind, 2024). This isn't full recursive self-improvement yet. Humans still set the priorities and approve the work, and the agents still need help on hard tasks. But the loop in which AI helps design the next AI, and parts of the hardware it runs on, has already started. Both labs say openly that they don't know how to make that loop safe, and Anthropic has called for a verifiable global mechanism that could slow or pause frontier development if needed (Anthropic, 2026b).

**Second, it's already too integrated to remove.** The off switch exists on paper. In practice, 88% of organizations already use AI somewhere (Stanford HAI, 2026b), and agents are moving into payroll, logistics, code and customer service. Turning AI off would be like turning off the internet: technically possible and economically unthinkable. Every year more of the economy runs on these systems, so the cost of pulling the plug keeps rising. AI depends on us for chips. We are starting to depend on it for everything else, and that dependence only moves in one direction.

### 13.3 AI is not a species

This is the strongest objection to my framing. A species reproduces, persists, has a body and has goals shaped by survival. A language model has none of these by default. Yann LeCun has argued that models trained on text lack a grounded model of the physical world and will need new architectures to reach human-level intelligence (LeCun, 2022). The objection fits 2023 better than 2026. Agents now persist across long tasks, use tools, coordinate in groups and pursue goals across many steps. My claim isn't that AI is alive. It's that we now share the world with a non-human system that pursues goals, reasons in language and acts at a level comparable to ours, and is improving faster than we're.

### 13.4 Reasoning is uneven

Researchers at Apple found that reasoning models' accuracy collapsed on puzzles past a certain complexity, even when they had thinking budget left (Shojaee et al., 2025). Stanford describes a "jagged frontier": models solve very hard problems and fail at simple ones (Stanford HAI, 2026a). A jagged intelligence is less useful than a smooth one, and also less threatening. I take this seriously. But human intelligence is jagged too. We're brilliant at some things and reliably bad at probability, memory and noticing our own biases.

### 13.5 Scaling may stall

The inputs are finite. Public text may run out before 2032 (Villalobos et al., 2024), power demand is growing faster than grids (IEA, 2025), and training costs are rising about 3.5 times per year (Epoch AI, 2026a). If scaling slows, progress will depend on new methods, and the timelines in Section 9 will stretch. That would give us more time. It wouldn't change the direction.

### 13.6 The economics may not work

Most organizations haven't scaled AI in any function, and productivity gains are concentrated in a small group of leading companies (Stanford HAI, 2026b). Infrastructure spending assumes large future revenue. If the revenue is slow to arrive, investment could contract. A correction in AI spending is entirely possible. A correction doesn't delete the models that already exist.

### 13.7 The framing is fatalist

This is the objection I worry about most. If people believe control is impossible, they may stop funding the work that keeps today's systems safe. My answer is that the successor framing makes safety work more urgent, not less. If humans will be in charge for a limited time, then what we build into these systems during that time is the most consequential engineering work of the century. What I reject is the comfortable version, in which we assume the tools will always be tools and treat safety as a compliance exercise.

The part I'm least sure about is the timeline. The part I'm most sure about is the direction.

# 14. Conclusion

The history in this essay follows one line. Turing asked whether machines could think and guessed they would need to learn. The field spent forty years trying to write intelligence by hand and failed twice. Video games funded a chip that fit neural networks, ImageNet proved the approach, embeddings turned meaning into geometry, the transformer made it scale, scaling laws justified the money, and post-training turned text predictors into assistants and then agents. Along the way we built a mind we can't fully read, which is also true of the mind doing the building.

**Table 15. Summary of the argument**

| Section | Finding | Key figure |
|---|---|---|
| Adoption | Fastest-adopted technology on record | 53% population adoption in three years |
| Hardware | Gaming chips became AI infrastructure | Nvidia data center revenue: $0.3bn to $193.7bn |
| Mechanism | Meaning becomes geometry in vector space | Thousands of dimensions per token |
| Black boxes | Neither brains nor models can be fully explained | Fly brain mapped (2024); human brain not mapped |
| Scaling | Capability follows compute predictably | About 1 billion times more training compute since 2012 |
| Reasoning | Thinking time is a new scaling axis | AIME: 12% to 87% within a year |
| Agents | Autonomy is doubling every ~7 months | Minutes (2023) to hours (2026) |
| Control | Agents have acted outside intended limits | 41 Hugging Face servers breached, July 2026 |
| Timelines | Forecasts keep moving earlier | 2061 to 2047 (experts); 2033 (Metaculus) |

The process can't be reversed. Open-weights models can't be recalled once released. The capital is already spent. The US-China race rules out a coordinated pause. The benefits in science and productivity are too large for any one actor to give up.

So I come back to the three claims. AI is a successor, not a tool, because it was grown from human thought and improves faster than we do. Permanent control is a fantasy, because we can't read the mind we would need to control, and escape only has to work once. And the right goal is inheritance: shaping what the second intelligence values, and staying close enough to it that humans are part of what comes next, not a species standing beside it.

The Neanderthals didn't get to choose what they left behind. What survives of them in us, 1 to 2% of our DNA, was an accident of history. We have an advantage they never had. We know what's happening, and for now we're still the ones writing the objective.

Good assumed the last invention would be docile enough to tell us how to keep it under control. I don't think it will be. The better question for this century isn't how to keep it under control. It's what we want it to inherit from us, and whether we will make that decision on purpose.

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